

6 D

Find driving force: $F_D - f = ma$

$$F_D = (3500 \times 0.35) + 3250 = 4425 \text{ N}$$

Power supplied by engine: $P = Fv$

$$P = 4425 \times 16 = 70800 \text{ W}$$

At maximum speed, driving force = frictional force.

Since power supplied is constant,

$$70800 = 3250v$$

$$v = 21.8 \text{ m s}^{-1}$$

7 D